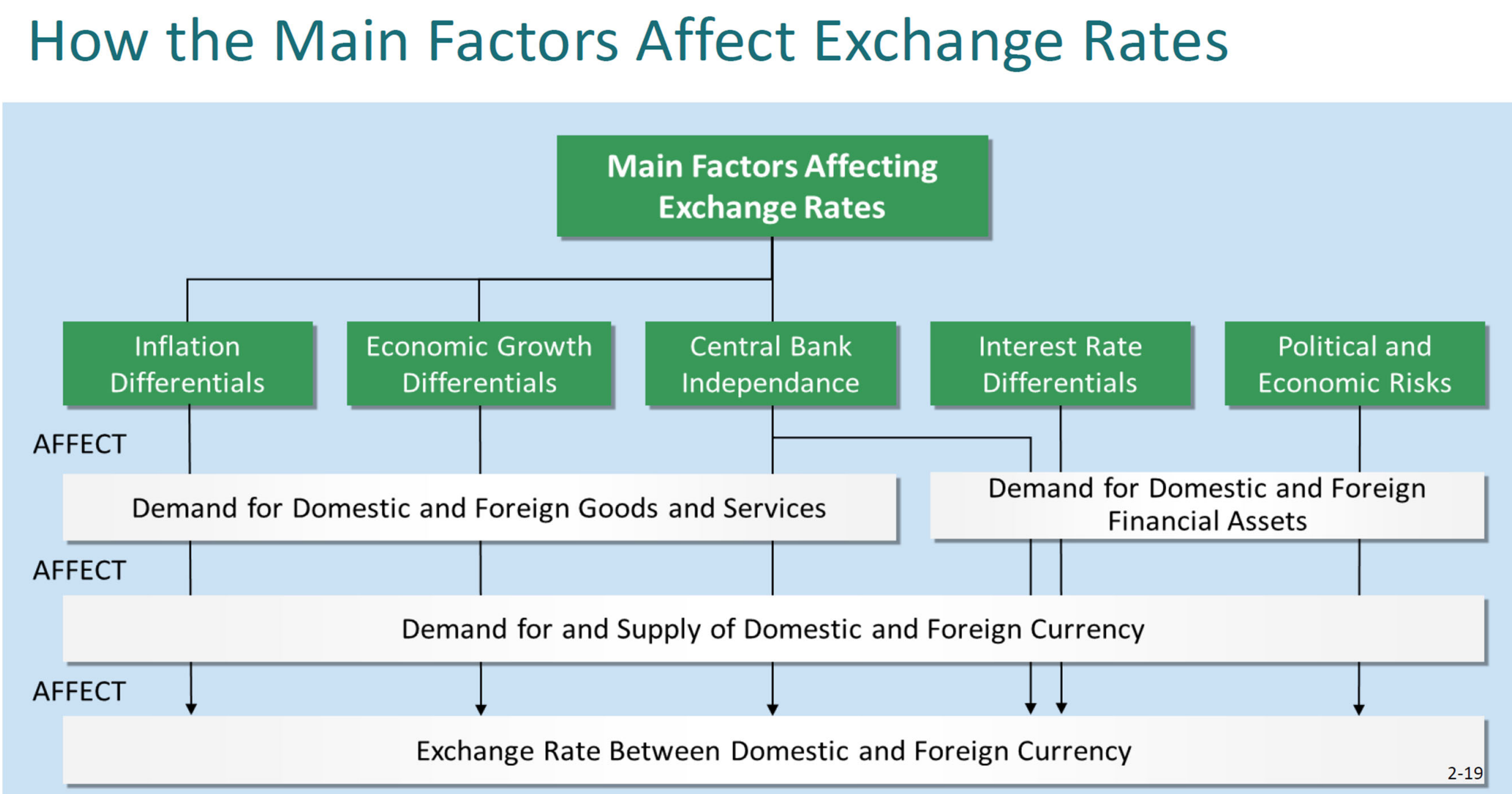
Chapter 4: Extra proofs and comments—Part 1

NOTE:

1. These notes do not contain any new examinable material.
2. They contain some elaborations on the lecture notes, with extra explanations and examples, and even a few mathematical derivations—which are not examinable.
3. I will refer to them in class.
4. However, students can ignore them if they don’t find them useful.

Slide 4-9: Recall Ch. 2, **Slide 2-19**:



Slide 4-8

Suppose an exchange rate is DKK 6.89/USD.

Which is the *price* of 1 DKK in USD?

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a) 6.89 DKK/USD b) $\frac{1}{6.89} = 0.145$ USD/DKK?

How else can we write this exchange rate?

a) 1 DKK = 6.89 USD? b) 1 USD = 6.89 DKK?

Slide 4-8

One share of NOVO is worth (DKK = Danish Krone): DKK 441.65.
The exchange rate is DKK 6.89/USD.
What is the price (of NOVO = NVO) in USD?
Is it 441.65×6.89, or 441.65/6.89?

One way to see it is to include the units:
DKK 441.65× DKK 6.89/USD—the overall units are DKK^2/USD.
DKK 441.64/ (DKK 6.89/USD)—the overall units are
DKK/(DKK/USD) = USD.
So, the correct answer is
DKK 441.65/DKK 6.89/USD = USD 64.10.

Alternatively, the *price* of 1 DKK in terms of USD is USD 1/6.89.

Slide 4-11: The Consumer Price Index (CPI)

On this slide, P_h and P_f can represent the price of an individual good, but in practice they are taken to represent the overall price level, or price index. These are typically proxied by Consumer Price Indices (CPI) in the home and foreign countries.

The growth rate of the CPI represents inflation. At date t: $\dfrac{P_t - P_{t-1}}{P_{t-1}} = i_t$.

(CPI measures changes in prices for a representative basket of consumption goods.)

Slide 4-15 (Relative PPP)

From Absolute PPP (slide 4-11), we saw that the following should hold:

$$P_h = e_0 P_f \text{ so that } e_0 = \frac{P_h}{P_f}.$$

If this holds at date 0 and continues to hold, then

at date *t*, we would expect

$$e_t = \frac{P_h(1+i_h)^t}{P_f(1+i_f)^t} = \frac{P_h}{P_f} \times \frac{(1+i_h)^t}{(1+i_f)^t} = e_0 \times \frac{(1+i_h)^t}{(1+i_f)^t}.$$

Slide 16

Suppose the current exchange rate is e_0.

Relative PPP says that in one year, (at $t = 1$)

$$e_t = e_0 \frac{(1+i_h)}{(1+i_f)} = e_0 \frac{(1+0.05)}{(1+0.01)} = e_0(1.0396),$$

which is an increase of about 4%.

Intuition: If US inflation is 4% higher than Japan's, the USD should depreciate by about 4% against the JPY so that US goods don't become persistently more expensive than Japanese goods once you account for the exchange rate.

Aside: a math result:

Suppose $\frac{1+x}{1+y}=1+a$ and $a\times y\approx 0$. Then, cross-multiplying gives

$$1+x=(1+a)(1+y)=1+a+y+a\times y\approx 1+a+y.$$

So, $a\approx x-y$. But $a=\frac{1+x}{1+y}-1$ by assumption, which gives

$$\frac{1+x}{1+y}-1\approx x-y.$$

Slide 4-18: For t = 1,

$$\frac{e_t-e_0}{e_0}=\frac{1}{e_0}\left(e_t-e_0\right)=\frac{1}{e_0}\left(e_0\times\frac{(1+i_h)}{(1+i_f)}-e_0\right)=\frac{(1+i_h)}{(1+i_f)}-1\approx i_h-i_f.$$

Slide 4-24: Real Rate of Interest, Nominal Rate of Interest, and Inflation

Example: A can of Coke costs p_0 = \$2.00 at time 0, and with inflation it is expected to cost $\tilde{p}_1$ = \$2.10 in one year, i.e., *the inflation rate* is

$$i = \frac{\tilde{p}_1 - p_0}{p_0} = 5\%.$$

The current 1-year *nominal interest rate* is r = 9%. If you buy 100 Cokes today, you will pay \$200. If you invest the \$200 for one year, you will have \$218 and since Coke will cost \$2.10 per can, you can buy 218/2.10 = 103.81 cans.

So, we say that the *real* interest rate is $a = 3.81\%$; the formula is $1+a=\dfrac{1+r}{1+i}$.

This is the Fisher Effect.

From our maths result, $a=\dfrac{1+r}{1+i}-1\approx r-i$.